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Revenue Cycle Management

Executive Summary

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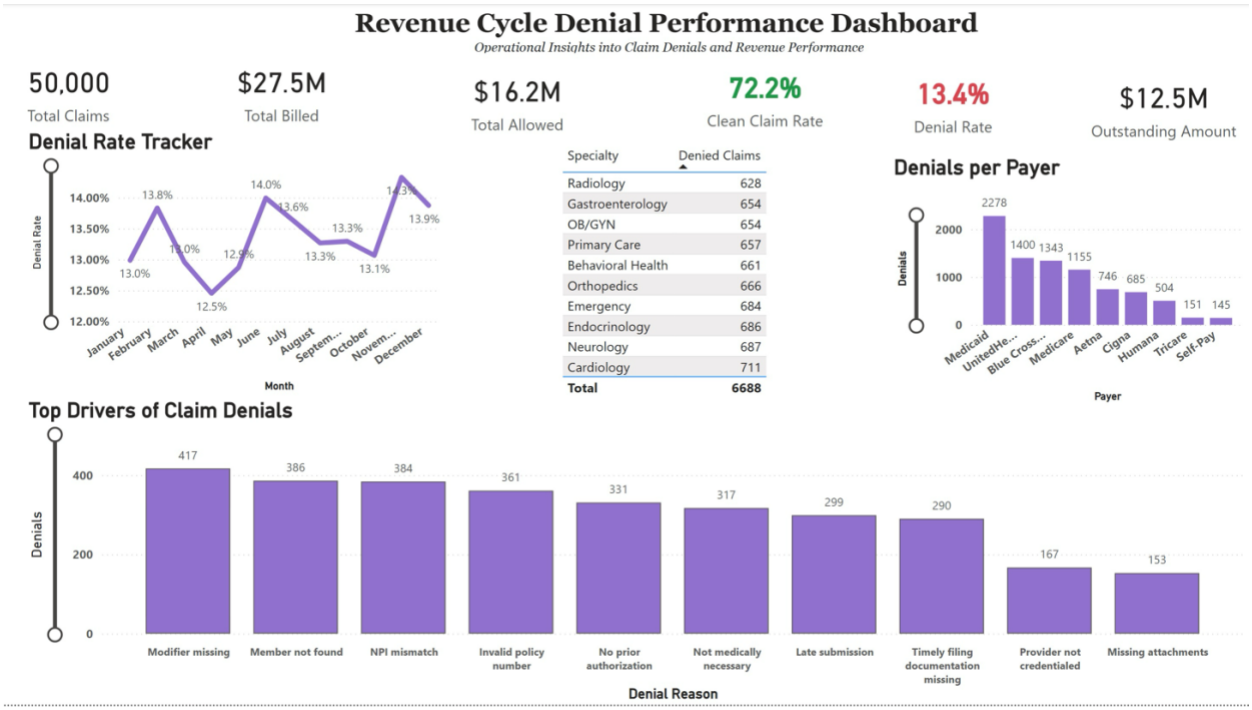
Revenue Cycle Management Independent Project



Executive Summary

This dashboard analyzes revenue cycle performance with a focus on identifying key drivers of claim denials and opportunities to improve financial outcomes. Across 50,000 claims and $27.5M billed, the denial rate sits at 13.4%, with a clean claim rate of 72.2%, indicating significant room for improvement in first-pass claim accuracy.

The analysis reveals that the leading causes of denials are operational in nature—primarily coding errors, missing modifiers, and member eligibility issues, rather than payer-side complexity alone. Denials are also unevenly distributed across payers and specialties, suggesting targeted intervention opportunities.

By addressing upstream workflow inefficiencies and strengthening validation processes, organizations can reduce denials, accelerate reimbursements, and improve overall revenue cycle performance.

1. Operational errors are the primary driver of denials

Top reasons:

- Modifier missing (417)
- Member not found (386)
- NPI mismatch (384)

These findings suggest denials are primarily driven by internal workflow inefficiencies rather than payer complexity.

2. Clean claim rate is underperforming

- 72.2% clean claim rate

Industry benchmark:

- ~85–95% (Rivet Health, n.d.)

There is a massive efficiency gap → rework → delays → lost cash flow

3. Denials are concentrated by payer

- Medicaid + UnitedHealthcare + BCBS dominate denials

Not all payers behave the same → targeted strategy needed

3 OPERATIONAL RECOMMENDATIONS

1. Strengthen coding accuracy through training + QA layers

- targeted training for entry-level coders
- implement pre-submission validation checks

Reduces:

- modifier missing
- NPI mismatch

2. Add front-end eligibility verification workflows

• automate member validation before submission

• integrate real-time eligibility checks

This reduces:

• “member not found” denials

3. Implement payer-specific denial management strategies

• analyze top denial patterns per payer

• build tailored submission rules

Outcome:

• fewer repeat denials

• faster reimbursement cycles

Documented Steps

1. Business Problem

Healthcare organizations lose revenue due to claim denials, often driven by preventable operational inefficiencies. This project analyzes denial patterns to identify root causes and recommend workflow improvements.

2. Data Overview

- Claims dataset (50,000 records)
- Payments dataset
- Key fields:
  - ClaimID
  - Payer
  - Denial Reason
  - Allowed Amount
  - Status

3. Data Model

A one-to-many relationship was established between Claims and Payments using ClaimID, ensuring accurate aggregation of financial metrics.

4. Key Measures (DAX / logic)

- Denial Rate = Denied Claims / Total Claims
- Clean Claim Rate = First-pass approvals / Total Claims
- Total Billed, Total Allowed, Outstanding Amount

5. Key Insights

- Operational errors are the primary driver of denials
- The clean claim rate is underperforming
- Denials are concentrated by the payer

6. Recommendations

- Strengthen coding accuracy through training + QA layers
- Add front-end eligibility verification workflows
- Implement payer-specific denial management strategies

7. Limitations

This analysis is based on simulated data and may not reflect real-world payer contracts or clinical complexity. Future iterations could incorporate claim timelines and denial resolution workflows.

This analysis demonstrates how targeted operational improvements can significantly reduce denials and improve revenue cycle efficiency at scale.

Reference

Rivet Health. (2024, October 1). A clean claim rate for your practice. Rivet. https://www.rivethealth.com/blog/a-clean-claim-rate-for-your-practice